TEXORA · SESSIONAL NOTES · STUDY SHEET

Practical Study on Computerized Embroidery Machine

The multi-head embroidery machine as a computer-controlled stitching system: heads, frames and drives, the digitized design file that steers them, and a verified production-time calculation.

AM-4202 · ADVANCED APPAREL MANUFACTURING-II SESSIONAL · STUDY 2 · LEVEL 4 TERM 2

1. Aim and place of the study

Aim: to identify the parts of a computerized multi-head embroidery machine, to follow a design from digitized file to stitched motif, and to calculate the machine time and output for a given design from its stitch count.

2. Materials, machines and specimens

Machine: multi-head computerized embroidery machine (the lab machine; head count and needle count per head recorded from its plate). **Materials:** garment panels or caps as specimens, cut-away and tear-away backing (stabilizer), embroidery threads (typically viscose or polyester, finer than sewing thread), bobbin thread, hoops/frames, the design file on USB, temporary spray adhesive, tweezers and small scissors.

3. How the machine is built

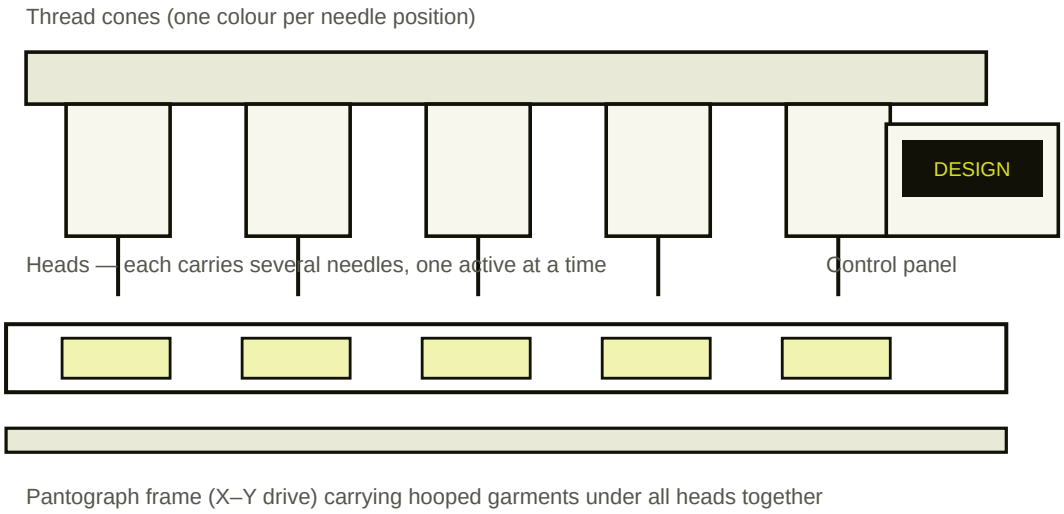


Figure 1 — Multi-head embroidery machine, schematic. All heads stitch the same design at the same moment; the frame, not the heads, moves in X–Y.

PART / SYSTEM	FUNCTION
Heads and needles	Each head holds several needles (commonly 9–15), each threaded with a different colour. Only one needle per head sews at a time; the head shifts sideways to bring the colour the design calls for over the needle bar.
Stitch formation	A lockstitch (301) formed with a rotary hook under the table, exactly as in Study 1 — embroidery is lockstitch sewing in which the fabric, not the feeding, draws the pattern.
Pantograph frame drive	Servo motors move the whole frame in X and Y between stitches, following the design file coordinate by coordinate; the needle simply goes up and down at the commanded point.
Hoops and frames	Clamp the garment panel and its backing flat and taut at each head position — registration and freedom from puckering are decided here, before any stitch is sewn.
Control unit and design file	The digitized design (stitch coordinates, colour changes, trims) is loaded from USB/network; the panel sets speed, assigns thread colours to needle numbers, and reports stitch count and progress.
Trimmers and thread-break sensors	Automatic trimmers cut top and bobbin threads at colour changes and design end; sensors stop the machine on a thread break so the operator can re-thread and back up a few stitches.

4. From artwork to stitches: the working sequence

1. **Digitizing (punching).** The artwork is converted in embroidery software into stitch data: areas are assigned stitch types — **running stitch** for outlines and detail, **satin stitch** for narrow columns and lettering, **tatami (fill) stitch** for broad areas — with stitch length, density and underlay set for the fabric. The digitizer, not the machine, decides how the design will look.
2. **Set-up.** Load the file; assign each design colour to a needle number and thread the heads to match; set the machine speed for the fabric and design.
3. **Hooping.** Backing and panel are hooped together, taut and square, positioned so the design lands at the marked placement point.
4. **Trace and trial.** Run the frame trace (the machine outlines the design area without sewing) to prove the design fits the hoop; sew one trial piece and inspect density, registration and puckering.
5. **Production run.** All heads sew together; the operator watches for thread breaks, re-threads and backs up over the break point so no gap remains.
6. **Finishing.** Remove the hoop, trim jump threads, tear or cut away the backing, and inspect against the approved sample.

5. Worked production calculation

The design stitched in this study contains **45,000 stitches** (read from the control panel's design data, which the digitizing software reports before a single stitch is sewn). The machine is set to **850 stitches per minute** — a normal working speed for a detailed design on a woven panel. Pure stitching time per piece:

$$\text{time} = 45{,}000 \div 850 = 52.94 \text{ min per piece per head}$$

Because every head sews the same design simultaneously, a **15-head** machine completes 15 pieces in that same 52.94 minutes, giving a stitching-rate output of $60 \div 52.94 \times 15 =$ **17.0 pieces per hour**. Hooping, unhooping and thread-break stops are *not* in this figure: at a realistic 85% running efficiency the line plans on about 14 pieces per hour, and it is this lower figure that belongs in a costing (compare Study 5, where the same efficiency logic is used for sewing lines).

> **READING THE NUMBERS**
>
> Stitch count is the embroidery equivalent of SMV: it is fixed by the digitizer's choices (a denser fill or a larger motif raises it directly), and it converts to money through machine time. Two designs of the same visual size can differ twofold in stitch count — which is why buyers approve the stitched sample, and why costing an embroidery design from its dimensions alone is meaningless.

Precautions

- Never reach into a head or under the frame while the machine is running; use the stop control first — the frame moves without warning between stitches.
- Prove placement with the trace function before the first stitch: a design sewn across a hoop edge breaks needles and ruins the garment.
- Keep backing matched to the fabric: an under-supported design puckers, and the fault is set-up, not machine adjustment.
- After a thread break, back up only a few stitches — backing up too far double-stitches a visible ridge into the design.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why does the frame move instead of the heads, and what does each head contribute?
2. Name the three basic stitch types a digitizer assigns, and where each is used.
3. A design of 28,000 stitches runs at 800 spm on a 12-head machine. Find the stitching time per batch of 12 and the pieces per hour.
4. What is backing for, and what fault appears when it is too weak for the design?

Answer key

1. Moving one frame in X–Y under all heads keeps every head mechanically simple (needle up/down plus colour selection) and guarantees all heads stitch identical pieces in register. Each head contributes one garment position and the set of thread colours on its needles.
2. Running stitch — outlines, fine detail, underlay travel. Satin stitch — narrow columns, borders and lettering, where stitches lie side by side across the column. Tatami/fill stitch — broad filled areas, built from rows of short stitches.
3. Time = 28,000 ÷ 800 = 35.0 min for 12 pieces (one per head). Output rate = 60 ÷ 35 × 12 = 20.6 pieces per hour (before efficiency losses).
4. Backing (stabilizer) supports the fabric against the pull of thousands of stitches so the design sews flat and in register. Too weak a backing lets the fabric draw in: puckering around the motif, distorted outlines and misaligned colour areas.